

(a) **Method inclusion and QC audit (17 candidates)**



7 excluded methods (audit)

●

raceid3

●

scAIDE

●

SCANER

●

MACE

●

CopyKAT

●

SCEVAN

●

scATOMIC

Output not cell-level

Failed at scale on Track A

Modality-incompatible (multi-omics)

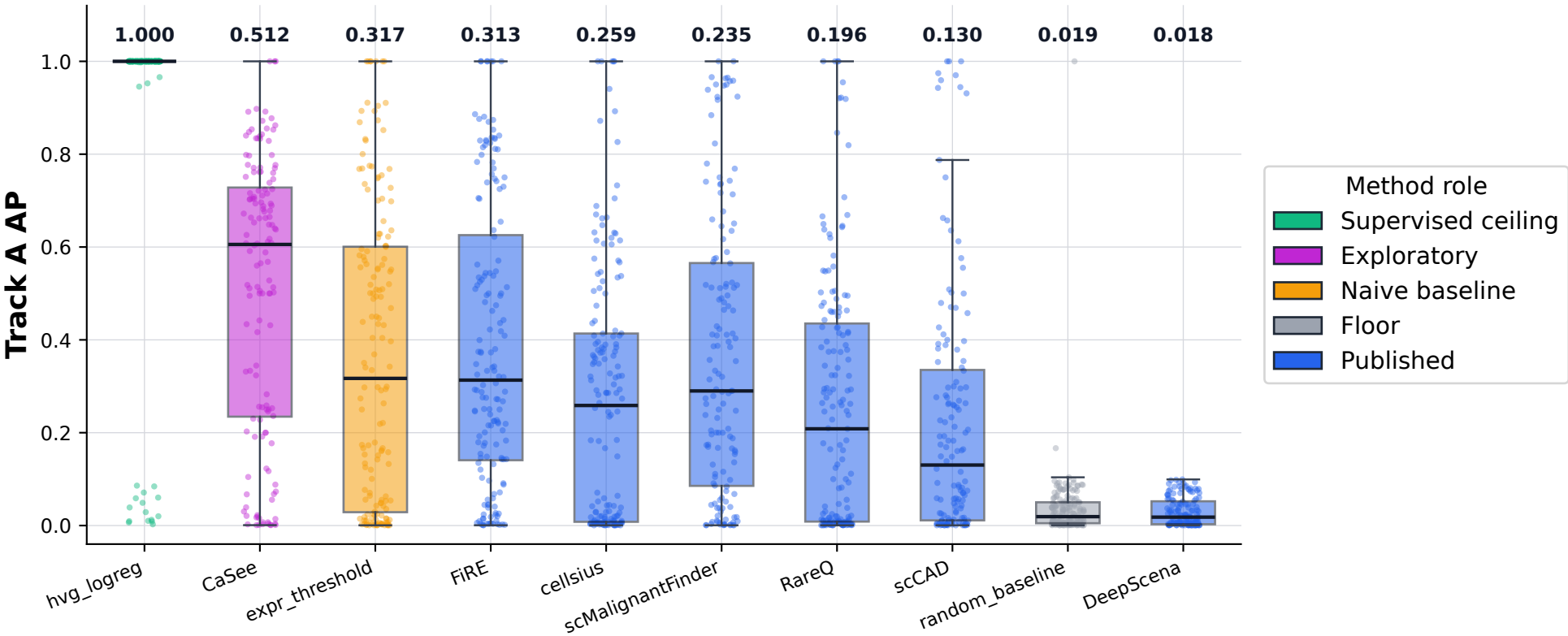
Returned degenerate constant outputs

CNV-only; no expression score

Tumour-only, not a ranker

Classifier, not a ranker

(b) **Track-A AP leaderboard**



(c) **Phase-11 multi-metric summary**

